Spark plug

A spark plug has two electrodes which are separated with each other. When a high tension current is applied between these two electrodes, it produces spark to ignite the air fuel mixture in the engine cylinder.

Working Principle

When the ignition switch is turned "ON", the current flows from the battery to the primary winding, through the contact breaker. The flowing current induces a magnetic field in the primary winding. As the contact breaker opens, the current collapses resulting in high tension induction in the secondary winding. This high tension generated in the secondary winding is transferred to the distributor by a cable. The distributor distributes high tension to the spark plug located in the individual cylinder as per the firing order. A spark is generated in the spark plug to initialise the combustion of fuel and air.

10.2.1.2 (a) Rotating Armature type Magento System

Magneto ignition system is a special type of ignition system with its own electric generator to provide the required necessary energy for the vehicle. A magneto, when rotated by the engine, is capable of producing a very high tension and doesn't need a battery as a source of external energy. It is mounted on the engine and replaces all components of the coil ignition system except the spark plug. Magneto ignition system can be either rotating armature type or rotating magneto type.

Construction

Permanent magnets are stationary and acting as the North and South poles. The armature consisting of the primary and secondary windings all rotate between the poles of a stationary magnet. Primary winding will be made up of thick wire with less no. of turns (150 to 300 turns) and the secondary winding is made up of thin wire with more No. of turns (15000 to 25000 turns). Contact breaker points and condenser are connected to the primary winding and spark plug is connected to the secondary winding.

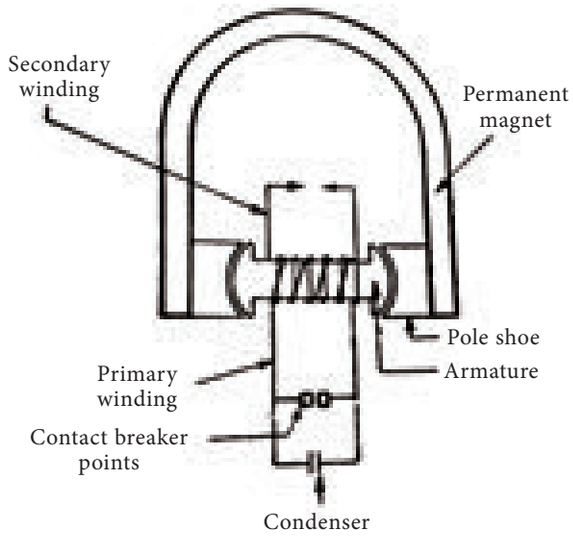


Figure 10.2.1.2 (a) Rotating armature type magnet ignition

Working

When the armature rotates, the flow of current is happening in primary winding and Contact breaker point in the closed condition. The magnetic field is induced in primary winding. When the contact breaker point open, the magnetic field in the primary winding collapses and by mutual induction principle, high tension current is induced in the secondary winding. This high tension current is applied to the spark

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